

Scott Okuma

Architect | Director | Manager

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Professional Summary

Accomplished spacecraft systems architect and software leader with nearly 15 years of experience guiding multidisciplinary teams in developing mission-critical flight software. Expert in embedded systems, real-time architectures, and avionics integration, with a proven track record of delivering reliable, scalable solutions for complex aerospace projects. Adept at technical leadership, team building, and aligning engineering with mission objectives.

Willing to relocate to: Thousand Oaks, CA - New York, NY - Los Angeles, CA

Authorized to work in the US for any employer

Work Experience

Spacecraft Systems Architect

Umbra Space-Goleta, CA

March 2025 to Present

- * Defined end-to-end architecture of spacecraft software systems, integrating real-time flight software, avionics, and ground command frameworks
- * Bridged hardware and software, ensuring cohesive integration across flight computers, data buses, and mission operations systems
- * Guided system-level software design, reliability strategies, and data architecture for mission-critical operations
- * Developed autonomy frameworks to support mission success from launch through end-of-life
- * Served as senior technical authority, aligning engineering execution with mission and business objectives
- * Led cross-functional teams to deliver robust, scalable software solutions for spacecraft systems
- * Established standards for software integration, verification, and validation processes

Director of Spacecraft Software

Umbra Space-Santa Barbara, CA

August 2021 to March 2025

- * Led all aspects of flight software development for spacecraft systems, overseeing architecture, development, and integration
- * Modular architecture layout for flight controller, GNC system, payload and radio systems.
- * Ensured reliability, safety, and mission success through disciplined engineering processes
- * Directed testing and validation of embedded flight software across multiple flight programs
- * Managed and improved nightly CI/CD efforts and test infrastructure
- * Fostered a culture of technical excellence and accountability within the software team

Flight Software Lead

Umbra Space-Santa Barbara, CA

June 2019 to September 2021

- * Developed and brought up a Zynq-based flight controller for small satellite command and control
- * Utilized Vivado/Vitis, C++, VHDL, Python, and embedded systems expertise to deliver robust solutions
- * Integrated DDR, microcontrollers, SPI, UART, Ethernet, and Linux kernels into flight systems
- * Led technical efforts in embedded systems and avionics software development

Electrical/Firmware Engineer

Raytheon-Goleta, CA

September 2017 to June 2019

- * Designed and implemented digital firmware for RF system controllers using VHDL/Verilog and Vivado
- * Customized Linux kernels and developed embedded systems with Petalinux and C/C++
- * Developed electronic attack VHDL modules, digital notch filters, and direct digital synthesis for defense applications
- * Conducted hardware checkout, schematic analysis, and DDR3 implementation for electronic modules
- * Utilized advanced test equipment including oscilloscopes, spectrum analyzers, and network analyzers

Electrical Engineer

NAWCWD-Point Mugu, CA

January 2015 to September 2017

- * Developed software and performed RF hardware repair, layout, and test planning for radar systems
- * Executed radar calibration and operations, utilizing waveform analysis and scenario programming
- * Created a DRFM jammer using FPGA, RF transceiver, and user interface for enhanced survivability
- * Programmed in Python, C++, and Matlab for signal processing and simulation tasks

Electrical/RF Engineer

CSC (Computer Sciences Corporation)-Point Mugu, CA

January 2012 to January 2015

- * Programmed and implemented automated test procedures for GaN RF amplifiers using Labview and GPIB interface
- * Analyzed test data in .csv format and generated plots using Octave for presentations
- * Verified thermal limits of equipment using environmental chambers and data logging software
- * Collaborated with technicians to construct final products and troubleshoot systems
- * Designed and built custom test fixtures for troubleshooting and software development

Satellite Communications Team Member for CSUN CubeSat1

Jet Propulsion Laboratory (JPL) / CSU Northridge

January 2014 to December 2014

- * Wrote and implemented radio communication test procedures for electrostatic sensitive devices
- * Assisted in antenna analysis and selection for CubeSat1
- * Used RealTerm and Logic Analyzer to examine satellite radio data
- * Created Bit Error Rate and Link Budget spreadsheets for mission planning
- * Ran satellite simulations with STK to determine optimal communication times
- * Developed Excel spreadsheets for link margin and data volume analysis

NOAA Radio Weather Receiver Team Member

CSU Northridge

February 2014 to June 2014

- * Designed RF amplifier for NOAA weather radio super-heterodyne receiver
- * Developed software-defined radio using MATLAB Simulink

Education

engineering (Bachelor's degree)

California State University Northridge

2012 to 2014

Skills

- CI/CD
- Embedded
- Flight software architecture
- Technical project management
- System integration (avionics, power, communications)
- C++
- Yocto
- microsoft office
- GitLab
- C
- Rust (programming language)
- Electrical engineering
- GitHub
- Embedded systems development
- Team Leadership and Mentoring
- FPGA Design
- Matlab
- Labview
- Uart
- Python
- Git
- Real-time operating systems (RTOS)
- Linux

Certifications and Licenses

Xilinx Embedded Design with Petalinux Tools

November 2018 to Present

Provides embedded systems developers with experience in creating an embedded Linux system targeting a Zynq® System on a Chip (SoC) processor and Zynq UltraScale+™ MPSoC processor development board using PetaLinux Tools.

The course provides experience with:

- Building the environment and booting the system using a Zynq SoC or Zynq UltraScale+ MPSoC design with PetaLinux Tools on the ARM® Cortex™-A9 or Cortex-A53 processor
- Using open-source embedded Linux components
- Configuring the Linux environment and network components
- Debugging embedded Linux platforms